

ALAMEDA, CA, USA

WMO: 994033

Lat: 37.772N

Lon: 122.298W

Elev: 2

StdP: 101.30

Time Zone: -8.00 (NAP)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.8	(c) 5.9	(d) -4.5	(e) 2.6	(f) 9.9	(g) -1.8	(h) 3.2	(i) 12.0	(j) 11.0	(k) 11.4	(l) 9.3	(m) 11.4	(n) 2.2	(o) 350	(p) 0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 9	(b) 6.8	(c) 27.4	(d) 17.8	(e) 25.2	(f) 17.2	(g) 23.3	(h) 16.7	(i) 19.2	(j) 25.9	(k) 18.4	(l) 24.6	(m) 17.8	(n) 23.2	(o) 3.1	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.1	20.5	16.1	11.4	19.9	15.4	10.9	19.6	54.3	26.0	51.9	24.7	49.8	23.1	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.0	(b) 7.8	(c) 6.9	(e) 2.8	(f) 31.9	(g) 1.9	(h) 2.8	(i) 1.4	(j) 34.0	(k) 0.3	(l) 35.6	(m) -0.8	(n) 37.2	(o) -2.2	(p) 39.3		
(a) 9.0	(b) 7.8	(c) 6.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	11.0	12.1	13.2	14.4	15.1	16.6	17.2	17.5	18.1	17.3	14.1	10.9
	DBStd	3.40	2.17	2.43	2.29	2.39	2.29	2.42	2.16	2.12	2.27	2.55	2.34	2.37
	HDD10.0	42	15	7	2	0	0	0	0	0	0	0	1	17
	HDD18.3	1380	228	174	158	121	104	64	49	40	29	51	129	231
	CDD10.0	1796	45	67	102	132	159	197	223	233	244	226	123	44
	CDD18.3	94	0	0	1	3	6	12	13	16	24	18	1	0
	CDH23.3	421	0	1	2	18	30	78	56	58	89	84	3	0
CDH26.7	98	0	0	0	3	6	26	14	13	17	19	0	0	
Wind	WSAvg	3.0	2.4	2.9	3.2	3.4	3.6	3.4	3.5	3.3	2.9	2.5	2.3	2.4
Precipitation	PrecAvg	454	93	70	71	31	6	3	1	1	6	32	69	67
	PrecMax	901	230	182	204	138	61	30	19	22	38	208	200	154
	PrecMin	184	7	3	0	0	0	0	0	0	0	0	3	0
	PrecStd	174	67	54	50	34	11	7	4	4	9	41	52	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	21.3	22.8	25.7	26.7	30.3	28.3	28.1	28.8	29.1	23.6	17.9
		MCWB	12.8	15.3	13.3	15.5	16.3	19.0	18.8	18.7	19.1	17.1	14.8	12.9
	2%	DB	16.8	18.8	19.8	22.4	22.9	25.2	24.5	24.9	25.8	25.6	20.8	16.4
		MCWB	11.7	14.2	14.1	15.4	16.4	17.5	17.6	17.8	17.6	17.0	14.3	12.3
	5%	DB	15.2	16.9	18.1	19.8	20.4	22.4	22.2	22.5	23.5	23.0	19.0	15.3
		MCWB	11.7	13.2	13.4	14.2	15.4	16.3	16.8	17.1	17.3	15.9	13.2	12.0
	10%	DB	14.0	15.5	16.6	18.0	18.6	20.5	20.6	20.9	21.6	21.0	17.3	14.4
		MCWB	11.2	12.2	12.8	13.6	14.2	15.3	16.0	16.4	16.5	15.3	12.8	11.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	16.4	16.4	17.7	18.0	19.9	19.7	20.1	20.2	19.2	17.2	15.1
		MCDB	15.9	20.4	20.9	23.9	23.3	29.8	28.3	25.7	25.8	27.5	20.0	16.3
	2%	WB	13.4	15.0	15.1	16.3	16.9	18.4	18.5	18.9	18.6	17.9	15.6	14.0
		MCDB	15.0	18.7	18.6	21.3	21.7	26.7	25.4	24.5	23.9	24.4	19.3	15.7
	5%	WB	12.5	13.9	14.4	15.5	16.2	17.2	17.6	18.1	17.8	17.0	14.6	13.2
		MCDB	14.3	16.9	17.4	19.6	20.6	23.6	22.8	23.1	22.9	22.4	18.4	15.0
	10%	WB	11.6	13.0	13.8	14.7	15.3	16.3	16.8	17.4	17.0	16.2	13.7	12.1
		MCDB	13.7	15.5	16.5	18.3	19.0	21.2	21.5	21.8	21.8	20.9	17.4	14.3
Mean Daily Temperature Range	5% DB	MDBR	5.8	5.9	6.1	6.4	5.9	6.7	6.1	6.1	6.8	7.2	6.5	5.4
		MCDBR	7.7	8.3	8.5	9.9	9.4	10.8	9.0	9.0	10.5	10.9	9.0	6.1
		MCWBR	4.8	5.0	4.7	5.2	4.9	5.8	4.5	4.9	5.3	5.4	5.3	4.6
	5% WB	MCDBR	5.7	7.6	7.4	9.0	8.8	11.2	9.3	9.4	9.8	10.1	8.3	4.9
		MCWBR	4.5	5.0	4.7	5.0	4.9	5.8	4.6	5.0	5.2	5.3	5.4	4.4
Clear-Sky Solar Irradiance	taub	0.324	0.329	0.341	0.359	0.364	0.359	0.368	0.348	0.347	0.338	0.323	0.325	
	taud	2.555	2.537	2.491	2.408	2.423	2.442	2.419	2.475	2.463	2.529	2.556	2.556	
	Ebn at Noon	864	905	923	920	917	917	906	920	904	881	857	837	
	Edh at Noon	76	87	100	115	115	113	115	106	101	86	75	72	
All-Sky Solar Radiation	RadAvg	2.38	3.32	4.69	6.14	7.08	7.73	7.32	6.52	5.53	4.07	2.81	2.08	
	RadStd	0.34	0.41	0.42	0.49	0.46	0.33	0.25	0.28	0.21	0.26	0.18	0.30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	N/A	N/A	-1.19	N/A	N/A	N/A	N/A	-110	+135	N/A

Nomenclature: See separate page